**Esercizio 3.8.** Si risolva, tramite l'algoritmo del simplesso primale, il seguente problema di programmazione lineare:

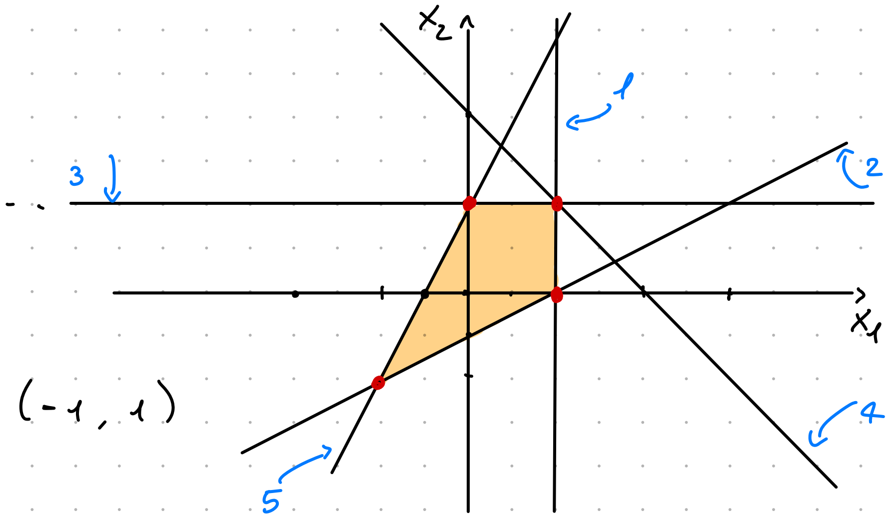
$$\max \quad x_2 - x_1$$

$$\begin{array}{rlcrl} x_1 \le 1 & 1 \leftarrow & \qquad & x_1 + x_2 \le 2 & 4 \\ 2x_2 + 1 \ge x_1 & 2 \leftarrow & & 2x_2 \le 4x_1 + 2 & 5 \\ -x_2 \ge -1 & 3 & & & \end{array}$$

Si parta dalla base ammissibile costituita dai vincoli $x_1 \le 1$ e $2x_2 + 1 \ge x_1$.

$$\max \; -x_1 + x_2$$

$$\begin{array}{lrcrcl} 1 & x_1 & & & \le & 1 \\ 2 & x_1 & - & 2x_2 & \le & 1 \\ 3 & & & x_2 & \le & 1 \\ 4 & x_1 & + & x_2 & \le & 2 \\ 5 & -4x_1 & + & 2x_2 & \le & 2 \end{array}$$




$$A = \begin{pmatrix} 1 & 0 \\ 1 & -2 \\ 0 & 1 \\ 1 & 1 \\ -4 & 2 \end{pmatrix} \qquad b = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 2 \end{pmatrix} \qquad c = (-1, 1)$$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad \det(A) = a \cdot d - b \cdot c$$

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\min \bar{y} b \quad t.c. \quad \bar{y} A = c, \quad \bar{y} \ge 0$$

$$B_1 = \{1, 2\}$$

$$A_{B_1}^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & -2 \end{pmatrix}^{-1} = -\frac{1}{2} \begin{pmatrix} -2 & 0 \\ -1 & 1 \end{pmatrix}$$

$$\bar{x}_1 = A_{B_1}^{-1} \cdot b_{B_1} = -\frac{1}{2} \begin{pmatrix} -2 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\bar{y}_1 = c \cdot A_{B_1}^{-1} = (-1; 1) \begin{pmatrix} 1 & 0 \\ 1/2 & -1/2 \end{pmatrix} = (\underbrace{-1/2}_{1}, \underbrace{-1/2}_{2})$$

$$h = 1$$

$$\xi = -(A_{B_1}^{-1})_1 = \begin{pmatrix} -1 \\ -1/2 \end{pmatrix} \qquad A_N \xi = \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ -4 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ -1/2 \end{pmatrix} = \begin{array}{c} \begin{pmatrix} \cancel{-1/2} \\ \cancel{-3/2} \\ 3 \end{pmatrix} \end{array} \begin{array}{c} 3 \\ 4 \\ 5 \end{array}$$

$K = 5$ → unica scelta

$$B_2 = \{2, 5\}$$

$$A_{B_2}^{-1} = \begin{pmatrix} 1 & -2 \\ -4 & 2 \end{pmatrix}^{-1} = -\frac{1}{6} \begin{pmatrix} 2 & 2 \\ 4 & 1 \end{pmatrix} = \begin{pmatrix} -1/3 & -1/3 \\ -2/3 & -1/6 \end{pmatrix}$$

$$\bar{x}_2 = A_{B_2}^{-1} \cdot b_{B_2} = \begin{pmatrix} -1/3 & -1/3 \\ -2/3 & -1/6 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = (-1; -1)$$

$$\bar{y}_2 = c \cdot A_{B_2}^{-1} = (-1; 1) \begin{pmatrix} -1/3 & -1/3 \\ -2/3 & -1/6 \end{pmatrix} = (\underbrace{-1/3}_{2}; \underbrace{1/6}_{5})$$

$h = 2$

$$\varepsilon = \begin{pmatrix} 1/3 \\ 2/3 \end{pmatrix} \qquad A_N \varepsilon = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1/3 \\ 2/3 \end{pmatrix} = \left( \frac{1/3}{1}, \frac{2/3}{3}, \frac{1}{4} \right)$$

$$\frac{b_1 - A_1 \bar{x}}{A_1 \varepsilon} = \frac{1 - (1 \; 0)\begin{pmatrix} -1 \\ -1 \end{pmatrix}}{1/3} = 6$$

$$\frac{b_3 - A_3 \bar{x}}{A_3 \varepsilon} = \frac{1 - (0 \; 1)\begin{pmatrix} -1 \\ -1 \end{pmatrix}}{2/3} = 3$$

$$\frac{b_4 - A_4 \bar{x}}{A_4 \varepsilon} = \frac{2 - (1 \; 1)\begin{pmatrix} -1 \\ -1 \end{pmatrix}}{1} = 4$$

$k = 3$

$$B_3 = \{3, 5\}$$

$$A_{B_3}^{-1} = \begin{pmatrix} 0 & 1 \\ -4 & 2 \end{pmatrix}^{-1} = \frac{1}{4} \begin{pmatrix} 2 & -1 \\ 4 & 0 \end{pmatrix} = \begin{pmatrix} 1/2 & -1/4 \\ 1 & 0 \end{pmatrix}$$

$$\bar{x} = A_{B_3}^{-1} b_{B_3} = \begin{pmatrix} 1/2 & -1/4 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = (0; 1)$$

$$y_{B_3} = (-1; 1) \begin{pmatrix} 1/2 & -1/4 \\ 1 & 0 \end{pmatrix} = (1/2; 1/4)$$

$$x^* = (0, 1)$$

$$y^* = (0, 0, 1/2, 0, 1/4)$$

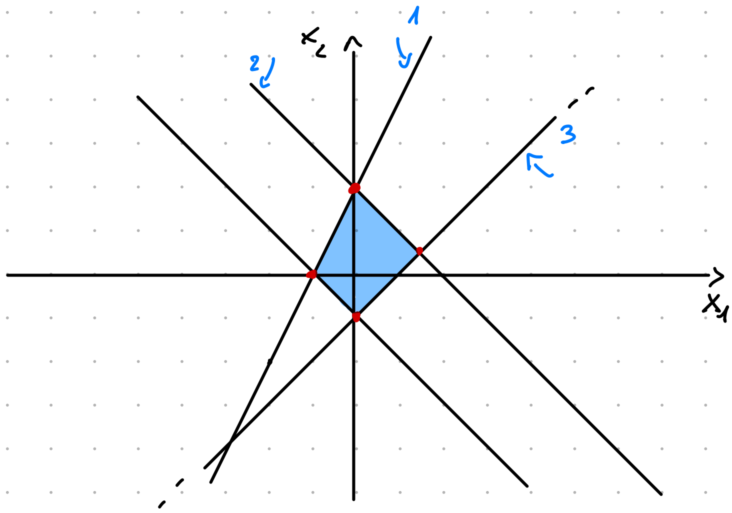
**Esercizio 3.11.** Si risolva, tramite l'algoritmo del simplesso primale, il seguente problema di programmazione lineare:

$$\min 2x_1 + x_2 + 3$$

$$2x_1 - x_2 \geq -2 \quad 1 \qquad\qquad x_2 \geq x_1 - 1 \quad 3$$

$$x_2 \leq -x_1 + 2 \quad 2 \qquad\qquad x_2 + x_1 \geq -1 \quad 4$$

Si parta dalla base ammissibile corrispondente ai vincoli $x_2 \geq x_1 - 1$ e $x_2 \leq -x_1 + 2$.




$$\begin{array}{lrcrcl} 1 & -2x_1 & + & x_2 & \leq & 2 \\ 2 & x_1 & + & x_2 & \leq & 2 \\ 3 & x_1 & - & x_2 & \leq & 1 \\ 4 & -x_1 & - & x_2 & \leq & 1 \end{array}$$

$$A = \begin{pmatrix} -2 & 1 \\ 1 & 1 \\ 1 & -1 \\ -1 & -1 \end{pmatrix} \quad b = \begin{pmatrix} 2 \\ 2 \\ 1 \\ 1 \end{pmatrix} \quad c = (-2, -1)$$

$$B_1 = \{2, 3\}$$

$$A_{B_1}^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1} = -\frac{1}{2}\begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}$$

$$\bar{x}_1 = A_{B_1}^{-1} b_{B_1} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = (3/2, 1/2)$$

$$\bar{y}_1 = c \cdot A_{B_1}^{-1} = (-2, -1)\begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} = (\underset{2}{\underline{-3/2}}, \underset{3}{\underline{-1/2}})$$

$$h = 2$$

$$\xi = \begin{pmatrix} -\frac{1}{2} \\ -1/2 \end{pmatrix} \qquad A_N \xi = \begin{pmatrix} -2 & 1 \\ -1 & -1 \end{pmatrix}\begin{pmatrix} -1/2 \\ -1/2 \end{pmatrix} = (1/2, 1)$$

$$\frac{b_1 - A_1 \bar{x}}{A_1 \xi} = \frac{2 - (-2, 1)\begin{pmatrix} 3/2 \\ 1/2 \end{pmatrix}}{(-2, 1)\begin{pmatrix} -1/2 \\ -1/2 \end{pmatrix}} = \frac{\frac{9}{2}}{\frac{1}{2}} = 9$$

$$\frac{b_4 - A_4 \bar{x}}{A_4 \xi} = \frac{1 - (-1, -1)\begin{pmatrix} 3/2 \\ 1/2 \end{pmatrix}}{(-1, -1)\begin{pmatrix} -1/2 \\ -1/2 \end{pmatrix}} = \frac{1+2}{1} = 3$$

$$K = 4$$

$$B_2 = \{3, 4\}$$

$$A_{B_2}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}^{-1} = -\frac{1}{2}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & -1/2 \end{pmatrix}$$

$$\bar{x} = A_{B_2}^{-1} b_{B_2} = \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & -1/2 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = (0, -1)$$

$$\bar{y} = c \, A_{B_2}^{-1} = (-2, -1)\begin{pmatrix} 1/2 & -1/2 \\ -1/2 & -1/2 \end{pmatrix} = (\underset{3}{\underline{-1/2}}, \underset{4}{\underline{3/2}})$$

$h = 3$

$\varepsilon = \begin{pmatrix} -1/2 \\ 1/2 \end{pmatrix}$ $\qquad A_N \varepsilon = \begin{pmatrix} -2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} -1/2 \\ 1/2 \end{pmatrix} = \left( \underset{1}{\underline{3/2}}, \underset{2}{\underline{0}} \right)$

$K = 1$

$B_3 = \{1, 4\}$

$A_{B_3}^{-1} = \begin{pmatrix} -2 & 1 \\ -1 & -1 \end{pmatrix}^{-1} = \frac{1}{3} \begin{pmatrix} -1 & -1 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{3} & -\frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{pmatrix}$

$\bar{x} = A_{B_3}^{-1} b_{B_3} = \begin{pmatrix} -1/3 & -1/3 \\ 1/3 & -2/3 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = (-1; 0)$

$\bar{y} = c \cdot A_{B_3}^{-1} = (-2, -1) \begin{pmatrix} -1/3 & -1/3 \\ 1/3 & -2/3 \end{pmatrix} = (1/3, 0)$

$x^* = (-1, 0)$

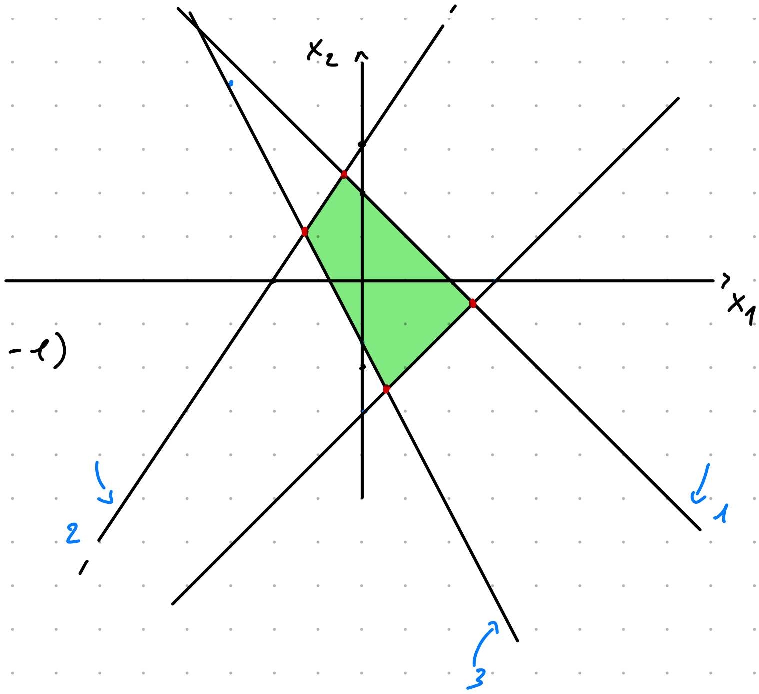
$y^* = (1/3, 0, 0, 0)$

**Esercizio 3.16.** Si risolva, tramite l'algoritmo del simplesso primale, il seguente problema di programmazione lineare:

$$\min x_2$$

$$
\begin{aligned}
x_1 + x_2 &\leq 2 & \quad 1\\
3x_1 - 2x_2 + 6 &\geq 0 & \quad 2\\
2x_2 + 4x_1 + 3 &\geq 0 & \quad 3\\
x_2 - x_1 + 3 &\geq 0 & \quad 4
\end{aligned}
$$

Si parta dalla base ammissibile corrispondente ai primi due vincoli.




$$
\begin{aligned}
1\quad & x_1 + x_2 \leq 2\\
2\quad & -3x_1 + 2x_2 \leq 6\\
3\quad & -4x_1 - 2x_2 \leq 3\\
4\quad & x_1 - x_2 \leq 3
\end{aligned}
$$

$$A = \begin{pmatrix} 1 & 1 \\ -3 & 2 \\ -4 & -2 \\ 1 & -1 \end{pmatrix} \qquad b = \begin{pmatrix} 2 \\ 6 \\ 3 \\ 3 \end{pmatrix} \qquad c = (0, -1)$$

$$B_1 = \{1, 2\}$$

$$A_{B_1}^{-1} = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 2/5 & -1/5 \\ 3/5 & 1/5 \end{pmatrix}$$

$$\bar{x} = \begin{pmatrix} 2/5 & -1/5 \\ 3/5 & 1/5 \end{pmatrix}\begin{pmatrix} 2 \\ 6 \end{pmatrix} = \left(-\frac{2}{5};\ \frac{12}{5}\right)$$

$$\bar{y}_{B_1} = (0, -1)\, A_B^{-1} = \left(\underbrace{-\frac{3}{5}}_{1},\ \underbrace{-\frac{1}{5}}_{2}\right)$$

$$h = 1$$

$$\xi = \begin{pmatrix} -2/5 \\ -3/5 \end{pmatrix} \qquad A_N \xi = \begin{pmatrix} -4 & -2 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} -2/5 \\ -3/5 \end{pmatrix} = \left(\underbrace{\frac{14}{5}}_{3},\ \underbrace{\frac{1}{5}}_{4}\right)$$

$$3 \qquad \frac{3 - (-4, -2)\begin{pmatrix} -2/5 \\ 12/5 \end{pmatrix}}{14/5} = \boxed{\frac{31}{14}}$$

$$4 \qquad \frac{3 - (1, -1)\begin{pmatrix} -2/5 \\ 1/5 \end{pmatrix}}{1/5} = 29$$

$$k = 3$$

$$B_2 = \{2, 3\} \qquad b_{B_2} = \begin{pmatrix} 6 \\ 3 \end{pmatrix}$$

$$A_B^{-1} = \begin{pmatrix} -3 & 2 \\ -4 & -2 \end{pmatrix}^{-1} = \begin{pmatrix} -1/7 & -1/7 \\ 2/7 & -3/14 \end{pmatrix}$$

$$\bar{x} = \begin{pmatrix} -1/7 & -1/7 \\ 2/7 & -3/14 \end{pmatrix} \begin{pmatrix} 6 \\ 3 \end{pmatrix} = \left( -\frac{9}{7}, \frac{15}{14} \right)$$

$$\bar{y}_{B_2} = (0, -1)\, A_{B_2}^{-1} = \left( -\frac{\frac{2}{7}}{2}, \frac{\frac{3}{14}}{3} \right)$$

$$h = 2 \qquad \varepsilon = \begin{pmatrix} 1/7 \\ -2/7 \end{pmatrix} \qquad A_N \varepsilon = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1/7 \\ -2/7 \end{pmatrix} = \left( -\frac{1/7}{1}, \frac{\frac{3}{7}}{4} \right)$$

$$k = 4$$

$$B_3 = \{ 3, 4 \}$$

$$A_{B_3}^{-1} = \begin{pmatrix} -4 & -2 \\ 1 & -1 \end{pmatrix}^{-1} = \frac{1}{6} \begin{pmatrix} -1 & 2 \\ -1 & -4 \end{pmatrix} = \begin{pmatrix} -1/6 & 1/3 \\ -1/6 & -2/3 \end{pmatrix}$$

$$\bar{x} = \begin{pmatrix} -1/6 & \frac{1}{3} \\ -1/6 & 2/3 \end{pmatrix} \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \left( 1/2, -\frac{5}{2} \right)$$

$$\bar{y} = (0, -1)\, A_{B_3}^{-1} = \left( \frac{1}{6}, \frac{2}{3} \right)$$

$$x^* = \left( 1/2, -\frac{5}{2} \right)$$

$$y^* = \left( 0, 0, 1/6, 2/3 \right)$$

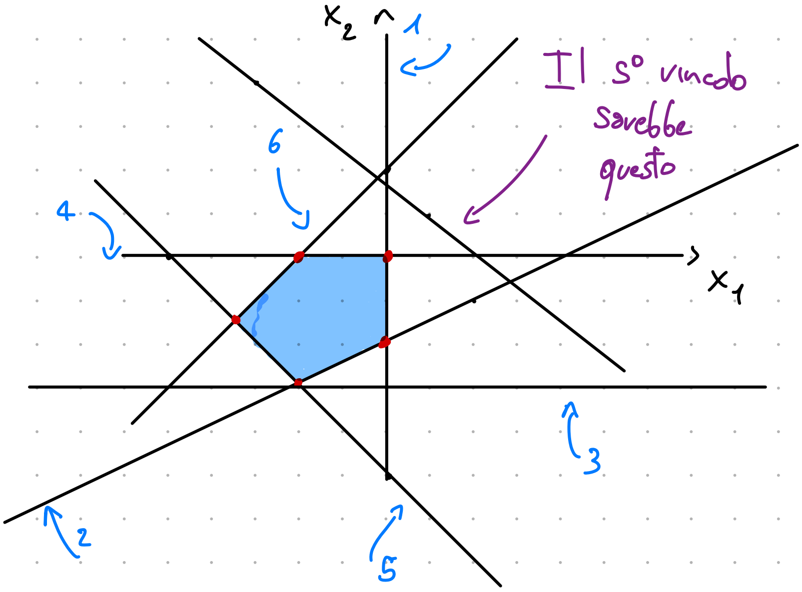
**Esercizio 3.26.** Si risolva, tramite l'algoritmo del simplesso primale, il seguente problema di programmazione lineare:

$$\min x_1 + 4x_2$$

$$
\begin{array}{llcll}
x_1 \le 0 & 1 & \qquad & \cancel{x_1 + x_2 \le 1} & 5 \\
2x_2 - x_1 + 4 \ge 0 & 2 & & x_1 + x_2 + 5 \ge 0 & 6 \\
x_2 + 3 \ge 0 & 3 & & x_1 - x_2 + 2 \ge 0 & 7 \\
x_2 \le 0 & 4 & & &
\end{array}
$$

Si parta dalla base ammissibile corrispondente ai vincoli ~~della prima riga~~. 1 e 4

$$
\begin{array}{lrcrcr}
1 & x_1 & & & \le & 0 \\
2 & x_1 & - & 2x_2 & \le & 4 \\
3 & & & -x_2 & \le & 3 \\
4 & & & x_2 & \le & 0 \\
5 & x_1 & + & x_2 & \le & 1 \\
6 & -x_1 & - & x_2 & \le & 5 \\
7 & -x_1 & + & x_2 & \le & 2
\end{array}
$$




$$A = \begin{pmatrix} 1 & 0 \\ 1 & -2 \\ 0 & -1 \\ 0 & 1 \\ -1 & -1 \\ -1 & 1 \end{pmatrix} \qquad b = \begin{pmatrix} 0 \\ 4 \\ 3 \\ 0 \\ 5 \\ 2 \end{pmatrix} \qquad c = (-1, -4)$$

$$B_1 = \{1, 4\}$$

$$A_{B_1}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\bar{x} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\bar{y} = (-1, -4)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = (\underset{1}{-1}, \underset{4}{-4})$$

$$h = 1$$

$$\xi = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \qquad A_N \xi = \begin{pmatrix} 1 & -2 \\ 0 & -1 \\ -1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} -1 \\ 0 \end{pmatrix} = (\underset{2}{-1}, \underset{3}{0}, \underset{5}{1}, \underset{6}{1})$$

$$\frac{b_5 - A_5 \bar{x}}{A_5 \xi} = \frac{5 - (-1, -1)\begin{pmatrix} 0 \\ 0 \end{pmatrix}}{(-1, -1)\begin{pmatrix} -1 \\ 0 \end{pmatrix}} = \frac{5}{1} = 5$$

$$\frac{b_6 - A_6 \bar{x}}{A_6 \xi} = \frac{2 - (-1, 1)\begin{pmatrix} 0 \\ 0 \end{pmatrix}}{(-1, 1)\begin{pmatrix} -1 \\ 0 \end{pmatrix}} = 2$$

$K = 6$

$B_2 = \{4, 6\}$

$A_{B_2}^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$

$\bar{x} = A_{B_2}^{-1} \, b_{B_2} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 2 \end{pmatrix} = (-2, 0)$

$\bar{y} = c \cdot A_{B_2}^{-1} = (-1, -4)\begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} = (\underset{4}{-5}, \underset{6}{1})$

$h = 4$

$\xi = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$ $\qquad A_N \xi = \begin{pmatrix} 1 & 0 \\ 1 & -2 \\ 0 & -1 \\ -1 & -1 \end{pmatrix}\begin{pmatrix} -1 \\ -1 \end{pmatrix} = (-1, \underset{2}{1}, \underset{3}{1}, \underset{5}{2})$

$\dfrac{b_2 - A_2 \bar{x}}{A_2 \xi} = \dfrac{4 - (1, -2)\begin{pmatrix} -2 \\ 0 \end{pmatrix}}{(1, -2)\begin{pmatrix} -1 \\ -1 \end{pmatrix}} = \dfrac{6}{1} = 6$

$\dfrac{b_3 - A_3 \bar{x}}{A_3 \xi} = \dfrac{3 - (0, -1)\begin{pmatrix} -2 \\ 0 \end{pmatrix}}{(0, -1)\begin{pmatrix} -1 \\ -1 \end{pmatrix}} = \dfrac{3}{1} = 3$

$\dfrac{b_5 - A_5 \bar{x}}{A_5 \xi} = \dfrac{5 - (-1, -1)\begin{pmatrix} -2 \\ 0 \end{pmatrix}}{(-1, -1)\begin{pmatrix} -1 \\ -1 \end{pmatrix}} = \dfrac{3}{2}$

$K = 5$

$B_3 = \{5, 6\}$

$A_{B_3}^{-1} = \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix}^{-1} = -\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$

$\bar{x} = A_{B_3}^{-1} \, b_{B_3} = \begin{pmatrix} -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}\begin{pmatrix} 5 \\ 2 \end{pmatrix} = \left(-\frac{7}{2}, -\frac{3}{2}\right)$

$\bar{y} = c \, A_{B_3}^{-1} = (-1, -4)\begin{pmatrix} -1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix} = \left(\underset{5}{\frac{5}{2}}, \underset{6}{-\frac{3}{2}}\right)$

$h = 6$

$\xi = \begin{pmatrix} +\frac{1}{2} \\ -1/2 \end{pmatrix}$ $\qquad A_N \xi = \begin{pmatrix} 1 & 0 \\ 1 & -2 \\ 0 & -1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix} = (\underset{1}{1/2}, \underset{2}{3/2}, \underset{3}{1/2}, \underset{4}{-1/2})$

$A_N \bar{x} = \begin{pmatrix} 1 & 0 \\ 1 & -2 \\ 0 & -1 \end{pmatrix}\left(-\frac{7}{2}, -\frac{3}{2}\right) = \begin{pmatrix} -\frac{7}{2} \\ -1/2 \\ 3/2 \end{pmatrix}$

$\dfrac{b_i - A_i \bar{x}}{A_i \xi} \longrightarrow \begin{pmatrix} 7 \\ 3 \\ 3 \end{pmatrix} \begin{matrix} 2 \\ 3 \end{matrix}$

Potrei scegliere entrambi
scelgo quello di indice minore

$K = 2$

$B_4 = \{2, 5\}$

$$A_{B_4}^{-1} = \begin{pmatrix} 1 & -2 \\ -1 & -1 \end{pmatrix}^{-1} = -\frac{1}{3} \begin{pmatrix} -1 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1/3 & -2/3 \\ -1/3 & -1/3 \end{pmatrix}$$

$$\bar{x} = A_{B_4}^{-1} b_{B_4} = \begin{pmatrix} 1/3 & -2/3 \\ -1/3 & -1/3 \end{pmatrix} \begin{pmatrix} 4 \\ 5 \end{pmatrix} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$$

$$\bar{y} = (-1, -4) \begin{pmatrix} 1/3 & -2/3 \\ -1/3 & -1/3 \end{pmatrix} = (1, 2)$$

$x^* = (-2, -3)$

$y^* = (0, -2, 0, 0, -3, 0)$

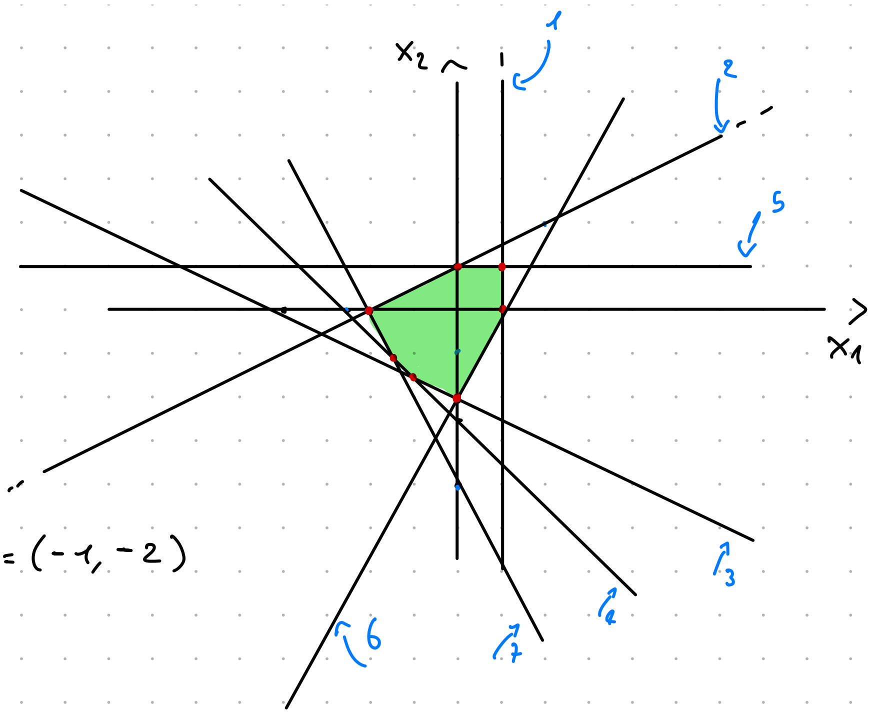
**Esercizio 3.29.** Si risolva, tramite l'algoritmo del simplesso primale, il seguente problema di programmazione lineare:

$$\cancel{\min x_1 + x_2} \quad \min x_1 + 2x_2$$

$$
\begin{aligned}
x_1 &\le 1 \quad 1 & \qquad x_2 &\le 1 \quad 5\\
2x_2 &\le 2 + x_1 \quad 2 & \qquad x_2 + 2 &\ge 2x_1 \quad 6\\
x_1 + 4 + 2x_2 &\ge 0 \quad 3 & \qquad 2x_1 + x_2 + 4 &\ge 0 \quad 7\\
2x_2 + 2x_1 + 5 &\ge 0 \quad 4 & &
\end{aligned}
$$

Si parta dalla base ammissibile corrispondente ai vincoli nella prima riga.

$$
\begin{array}{lrcrl}
1 & x_1 & & & \le 1\\
2 & -x_1 & + & 2x_2 & \le 2\\
3 & -x_1 & - & 2x_2 & \le 4\\
4 & -2x_1 & - & 2x_2 & \le 5\\
5 & & & x_2 & \le 1\\
6 & 2x_1 & - & x_2 & \le 2\\
7 & -2x_1 & - & x_2 & \le 4
\end{array}
$$




$$A = \begin{pmatrix} 1 & 0\\ -1 & 2\\ -1 & -2\\ -2 & -2\\ 0 & 1\\ 2 & -1\\ -2 & -1 \end{pmatrix} \qquad b = \begin{pmatrix} 1\\ 2\\ 4\\ 5\\ 1\\ 2\\ 4 \end{pmatrix} \qquad c = (-1, -2)$$

$$B_1 = \{1, 5\} \qquad b_B = \begin{pmatrix} 1\\ 1 \end{pmatrix}$$

$$A_{B_1} = \begin{pmatrix} 1 & 0\\ 0 & 1 \end{pmatrix} \qquad A_{B_1}^{-1} = A_{B_1}$$

$$\bar{x} = \begin{pmatrix} 1 & 0\\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1\\ 1 \end{pmatrix} = (1, 1) \qquad \bar{y}_{B_1} = (-1, -2) \begin{pmatrix} 1 & 0\\ 0 & 1 \end{pmatrix} = (\underset{1}{-1}, \underset{5}{-2})$$

$$h = 1 \qquad \xi = \begin{pmatrix} -1\\ 0 \end{pmatrix} \qquad A_N \xi = \begin{pmatrix} -1 & 2\\ -1 & -2\\ -2 & -2\\ 2 & -1\\ -2 & -1 \end{pmatrix} \begin{pmatrix} -1\\ 0 \end{pmatrix} = (\underset{2}{1}, \underset{3}{1}, \underset{4}{2}, \underset{6}{-2}, \underset{7}{2})$$

$$A_N \bar{x} = \begin{pmatrix} -1 & 2\\ -1 & -2\\ -2 & -2\\ -2 & -1 \end{pmatrix} \begin{pmatrix} 1\\ 1 \end{pmatrix} = (1, -3, -4, -3)$$

$$\frac{b_i - A_i \bar{x}}{A_i \xi} \longrightarrow \begin{pmatrix} 1\\ 7\\ 9/2\\ 7/2 \end{pmatrix}$$

$K = 2$

$B_2 = \{2, 5\}$

$A_{B_2}^{-1} = \begin{pmatrix} -1 & 2 \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} -1 & 2 \\ 0 & 1 \end{pmatrix}$

$\bar{x} = \begin{pmatrix} -1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = (0, 1)$

$y_B = (-1, -2) \begin{pmatrix} -1 & 2 \\ 0 & 1 \end{pmatrix} = (\underset{2}{1}, \underset{5}{-4})$

$h = 5$

$\varepsilon = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$ $\qquad A_N \varepsilon = \begin{pmatrix} 1 & 0 \\ -1 & -2 \\ -2 & -2 \\ 2 & -1 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ -1 \end{pmatrix} = (\cancel{-2}, \underset{3}{4}, \underset{4}{6}, \cancel{-3}, \underset{7}{5})$

$A_N \bar{x} = \begin{pmatrix} -1 & -2 \\ -2 & -2 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = (-2, -2, -1)$

$\dfrac{b_i - A_i \bar{x}}{A_i \varepsilon} \longrightarrow \begin{pmatrix} 3/2 \\ 7/6 \\ 1 \end{pmatrix} \hookleftarrow 7$

$K = 7$

$B_3 = \{2, 7\}$

$A_B^{-1} = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix}^{-1} = \frac{1}{5} \begin{pmatrix} -1 & -2 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} -1/5 & -2/5 \\ 2/5 & -1/5 \end{pmatrix}$

$\bar{x} = \begin{pmatrix} -1/5 & -2/5 \\ 2/5 & -1/5 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = (-2, 0)$

$\bar{y}_{B_3} = (-1, -2) \begin{pmatrix} -1/5 & -2/5 \\ 2/5 & -1/5 \end{pmatrix} = \left(-\frac{3}{5}, \frac{4}{5}\right)$

$h = 2$

$\varepsilon = \begin{pmatrix} 1/5 \\ -2/5 \end{pmatrix}$ $\qquad A_N \varepsilon = \begin{pmatrix} 1 & 0 \\ -1 & -2 \\ -2 & -2 \\ 0 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1/5 \\ -2/5 \end{pmatrix} = (\underset{1}{1/5}, \underset{3}{1/5}, \underset{4}{2/5}, \underset{5}{-2/5}, \underset{6}{4/5})$

$A_N \bar{x} = \begin{pmatrix} 1 & 0 \\ -1 & -2 \\ -2 & -2 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ 0 \end{pmatrix} = (-2, 2, 4, -4)$

$$\frac{b_i - A_i \bar{x}}{A_i \varepsilon} \longrightarrow \begin{pmatrix} 15 \\ 10 \\ 5/2 \\ \frac{30}{4} \end{pmatrix} \; 4 \qquad K = 4$$

$$B_4 = \{4, 7\}$$

$$A_{B_4}^{-1} = \begin{pmatrix} -2 & -2 \\ -2 & -1 \end{pmatrix}^{-1} = -\frac{1}{2}\begin{pmatrix} -1 & 2 \\ 2 & -2 \end{pmatrix} = \begin{pmatrix} 1/2 & -1 \\ -1 & 1 \end{pmatrix}$$

$$\bar{x} = \begin{pmatrix} 1/2 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 5 \\ 4 \end{pmatrix} = \begin{pmatrix} -3/2 \\ -1 \end{pmatrix}$$

$$\bar{y} = (-1, -2)\begin{pmatrix} 1/2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{array}{cc} (3/2, & -1) \\ \bar{4} & \bar{7} \end{array}$$

$$h = 7$$

$$\varepsilon = \begin{pmatrix} +1 \\ -1 \end{pmatrix} \qquad A_N \varepsilon = \begin{pmatrix} 1 & 0 \\ -1 & 2 \\ -1 & -2 \\ 0 & 1 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{array}{ccccc} (1, & -3, & 1, & -1, & 3) \\ 1 & 2 & 3 & 5 & 6 \end{array}$$

$$A_N \bar{x} = \begin{pmatrix} 1 & 0 \\ -1 & -2 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} -3/2 \\ -1 \end{pmatrix} = \left(-\frac{3}{2}, \frac{7}{2}, -2\right)$$

$$\frac{b_i - A_i \bar{x}}{A_i \varepsilon} \longrightarrow \begin{pmatrix} 5/2 \\ 1/2 \\ 4/3 \end{pmatrix} \; 3$$

$$K = 3$$

$$B_5 = \{3, 4\}$$

$$A_{B_5}^{-1} = \begin{pmatrix} -1 & -2 \\ -2 & -2 \end{pmatrix}^{-1} = -\frac{1}{2}\begin{pmatrix} -2 & 2 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & \frac{1}{2} \end{pmatrix}$$

$$\bar{x} = A_{B_5}^{-1} b_{B_5} = \begin{pmatrix} 1 & -1 \\ -1 & \frac{1}{2} \end{pmatrix}\begin{pmatrix} 4 \\ 5 \end{pmatrix} = (-1, -3/2)$$

$$\bar{y} = (-1, -2)\begin{pmatrix} 1 & -1 \\ -1 & \frac{1}{2} \end{pmatrix} = (1, 0)$$

$$x^* = \left(-1, -\frac{3}{2}\right)$$

$$y^* = (0, 0, 1, 0, 0, 0, 0)$$